

Keith's Notes for MA1200 PALSI: Differentiation

This document is primarily intended for the uses of students registered for PALSI at CityUHK, and participated in Keith TSE's section.

1 Introduction

These notes cover the key topics from MA1200 Calculus and Basic Linear Algebra I, including limits and differentiation.

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2 Limits

2.1 Motivation of Limit

Consider the function $f(x) = \frac{x^3+2x-3}{x^2-1}$. The function is not defined at $x = 1$ since the denominator is zero. However, we can investigate the value of $f(x)$ as x approaches 1.

From a table of values near $x = 1$, $f(x)$ approaches 2.5. This introduces the concept of limits.

2.2 Definition of Limit

Intuitively, $\lim_{x \rightarrow c} f(x) = L$ means that $f(x)$ gets arbitrarily close to L as x gets sufficiently close to c .

2.3 Limit of Continuous Functions

If $f(x)$ is continuous at c (no breaks or jumps), then $\lim_{x \rightarrow c} f(x) = f(c)$.

Examples of continuous functions: constants, polynomials, exponential functions, trigonometric functions like $\cos x$, and square roots.

2.4 Examples Where Limit May Not Exist

Jump discontinuity: Limit does not exist if left and right limits differ.

Infinite limit: Limit is not a real number if $f(x)$ tends to infinity.

Oscillation: Limit does not exist if $f(x)$ oscillates without approaching a value.

2.5 One-Sided Limits

Left-hand limit: $\lim_{x \rightarrow c^-} f(x) = L$ if $f(x)$ approaches L as x approaches c from the left.

Right-hand limit: $\lim_{x \rightarrow c^+} f(x) = L$ if from the right.

The two-sided limit exists if and only if both one-sided limits exist and are equal.

2.6 Properties of Limits

Assume $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$.

$$\lim_{x \rightarrow c} kf(x) = kL$$

$$\lim_{x \rightarrow c} [f(x) \pm g(x)] = L \pm M$$

$$\lim_{x \rightarrow c} [f(x) \cdot g(x)] = L \cdot M$$

$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}$ if $M \neq 0$
 $\lim_{x \rightarrow c} [f(x)]^a = L^a$ if defined.

2.7 Algebraic Tricks for Limits

Factorization and cancellation for indeterminate forms like $0/0$.

Rationalization for square roots.

Trigonometric identities like product-to-sum formulas.

2.8 Sandwich Theorem

If $g(x) \leq f(x) \leq h(x)$ and $\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L$, then $\lim_{x \rightarrow c} f(x) = L$.

Useful for limits like $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$.

2.9 Limits at Infinity

$\lim_{x \rightarrow \infty} f(x) = L$ if $f(x)$ approaches L as x becomes large.

Useful facts:

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

For $a > 1$, $\lim_{x \rightarrow \infty} a^x = \infty$, $\lim_{x \rightarrow -\infty} a^x = 0$

For $0 < a < 1$, opposite.

Divide by highest power for rational functions.

3 Continuity

3.1 Definition of Continuous Function

A function $f(x)$ is continuous at c if $\lim_{x \rightarrow c} f(x) = f(c)$.

Continuous on domain if continuous at every point in domain.

3.2 Mathematical Definition

f is continuous at a if $\lim_{x \rightarrow a} f(x)$ exists and equals $f(a)$.

3.3 Notes on Continuity

Most elementary functions (polynomials, exponentials, trig, roots) are continuous on their domains.

To check continuity at a : Compute $f(a)$.

Compute $\lim_{x \rightarrow a} f(x)$ (use one-sided if needed).

Compare.

3.4 Properties of Continuous Functions

Sum, difference, product, quotient (denominator $\neq 0$) of continuous functions are continuous. Composition of continuous functions is continuous.

3.5 Intermediate Value Theorem

If f continuous on $[a, b]$, and k between $f(a)$ and $f(b)$, then exists c in (a, b) with $f(c)=k$.

Useful for proving existence of roots.

3.6 Method of Bisection

Root-finding using IVT to narrow intervals.

4 Introduction to Differentiation

4.1 Why Differentiation?

Example: HK population growth from 1931 (864,117) to 1999 (6,840,000).

Average growth: $(6,840,000 - 864,117)/(1999 - 1931) \approx 87,881/\text{year}$.

But growth varies due to conditions (wars, economy).

To estimate at specific time, shorten interval, limit as $\Delta t \rightarrow 0$: rate of change.

4.2 Rate of Change

The instantaneous rate of change is $\lim_{\Delta t \rightarrow 0} \frac{P(t+\Delta t) - P(t)}{\Delta t}$.

This is the derivative.

5 More on Differentiation

5.1 Implicit Differentiation

For equations not solved for y, differentiate both sides w.r.t. x, solve for dy/dx.

Example: For $x^2 + y^2 = 1$, $2x + 2y \, dy/dx = 0$, $dy/dx = -x/y$.

5.2 Differentiation of Inverse Functions

For $y = f^{-1}(x)$, $\frac{dy}{dx} = \frac{1}{df/dy \text{ at } y=f^{-1}(x)}$.

Examples:

$$\begin{aligned}\frac{d}{dx} \arcsin x &= \frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx} \arccos x &= -\frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx} \arctan x &= \frac{1}{1+x^2} \\ d/dx \ln x &= 1/x\end{aligned}$$

5.3 Logarithmic Differentiation

For $y = [f(x)]^{g(x)}$, $\ln y = g(x) \ln f(x)$, differentiate: $(1/y) \frac{dy}{dx} = \dots$

Useful for products/powers.

5.4 Differentiation of Parametric Equations

For $x = f(t)$, $y = g(t)$, $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$, if $\frac{dx}{dt} \neq 0$.

Higher-order: $\frac{d^2 y}{dx^2} = \frac{d/dt(dy/dx)}{\frac{dx}{dt}}$.

5.5 Higher-Order Derivatives

nth derivative: differentiate n times.

Notation: $f^{(n)}(x)$, $\frac{d^n y}{dx^n}$.

Leibniz rule for product: $(fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}$.

Useful for products like $x^n \sin x$.

Partial fractions for rational functions to simplify differentiation.

Product-to-sum for trig products.

6 Applications of Differentiation

6.1 Approximation of Functions - Taylor Series

Taylor series: approximation using higher derivatives.

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \dots$$

Application of higher-order derivatives.

6.2 Optimization

Find max/min using first and second derivatives. Critical points: $f'(x)=0$.

Second derivative test: $f'' < 0$ min, > 0 max.

6.3 Computing Limits: L'Hôpital's Rule

For $0/0$ or ∞/∞ forms, $\lim \frac{f}{g} = \lim \frac{f'}{g'}$ if exists.

6.4 Root-Finding Algorithm: Newton's Method

$x_{n+1} = x_n - f(x_n)/f'(x_n)$ Converges to root.